Instructions

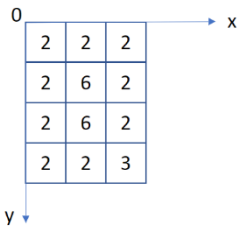
This lab has three parts. For each part, implement the outlined functions in lab4.py and run them in the notebook lab4.ipynb. Exploratory questions are embedded directly at the end of the sections; write your answers directly into the notebook. Expected outputs are supplied as reference, but your outputs do not need to match exactly.

Upload to Canvas your completed lab4.py and lab4.ipynb by zipping them into a file named AXXX1_AXXX2.zip, where AXXX is the student number of the group members. Submit one file per group. Missing files, incorrectly formatted code that does not run, etc. will be penalized.

Tanuj is the TA in charge of this lab. Please post any questions onto the Canvas forum or attend one of the FAQ sessions during the regularly scheduled Lab session on 13.04 or 15.04.

Objective

This lab covers keypoint detection and feature descriptors. We will explore the Harris Corner Detector, a simplified version of the SIFT descriptor and two applications for keypoint matching: image stitching and symmetry detection.



Unfortunately we need to change the coordinate convention (x- and y-axis are swapped compared to Lab 2, see Figure on the right) to stay consistent with the conventions of the OpenCV functions which are used in this lab.

Part 1: Keypoint Detection, Description, and Matching (40%)

This part covers foundations of keypoint descriptors and matching which will be used in later parts. We provide the wrapper `describe_keypoints` which calls either the naïve descriptor or the simplified version of SIFT.

Keypoint detection (7%)

- `harris_corners()` detects Harris corner keypoints. For each pixel in an image,

1. Compute image gradient I_x, I_y and compose the second moment matrix H. Note that the elements A, B, and C of matrix H are obtained by summing the elements in some window W:

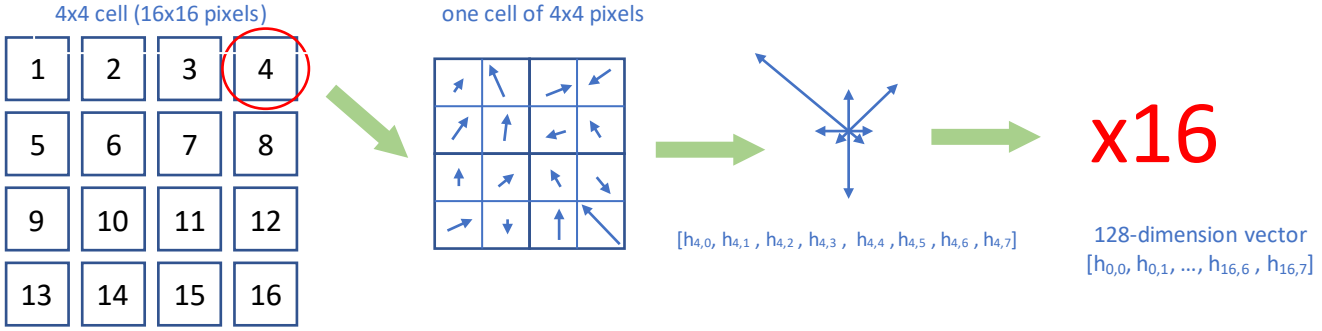
   i. $H = \begin{bmatrix} A & B \\ B & C \end{bmatrix}$ $A = \sum_{(x,y)\in W} I_x^2$ $B = \sum_{(x,y)\in W} I_x I_y$ $C = \sum_{(x,y)\in W} I_y^2$

   Consider an efficient implementation of the window-wise summation based on convolution (i.e. do not use for-loops). Weight the elements uniformly in the window for the summation. Hint: Consider using built-in functions `skimage.filters.sobel_v`, `sobel_h` and `scipy.ndimage.filters.convolve`.
2. Compute the corner response $R = det(H) - k\left(tr(H)\right)^2$ with $k = 0.04$.
3. Find local maxima using the provided non-maximum suppression; the local maxima in the response map corresponds to the keypoints.

Keypoint description (15%)

- `naive_descriptor()` computes a local descriptor based on the normalized pixel intensity, i.e. $F(x,y) = \frac{I(x,y)-\mu}{\sigma+0.0001}$. $I(x,y)$ is the intensity at a keypoint located at (x,y) and μ and σ are the mean and standard deviation of the pixel intensities in a patch of 5x5 centered at (x,y). The 0.0001 term in the numerator improves numerical stability.
- `simple_sift()` is a simplified version of the full SIFT descriptor which is not rotationally invariant. For each keypoint, take a 16×16 patch of pixels around the keypoint and compute the image gradient magnitude and orientation at each pixel. For efficiency, consider

precomputing the image gradients for the entire image in advance. Divide the patch into a 4×4 grid of cells, where each cell has $4 \times 4 = 16$ pixels (see Figure 1a).



(a) 16x16 patch is partitioned into a 4x4 array of cells; each cell is 4x4 pixels. Note the numbers do not correspond to indices.

(b) Histogram of gradient orientations from each cell; the 8 bin histogram forms an 8-dim. feature vector. Concatenating this vector from all 16 cells in the patch forms a 128d feature.

Figure 1. Schematic visualization of estimating our simplified SIFT descriptor.

For each cell, create a histogram of the gradient orientations with 8 bins of 45° each (e.g. 0° to 44° for bin 0, 45° to 89° for bin 1, etc.). Each pixel's contribution to the histogram is weighted by the pixel's gradient magnitude and the weight from a Gaussian kernel of 16×16 centered on the keypoint coordinate. Use the provided `make_gaussian_kernel`. Concatenate the histograms from each cell to form a 4×4×8=128-dimension vector (see Figure 1b). Normalize the vector to unit length by dividing by the vector's magnitude [link].

Keypoint matching (8%)

- `top_k_matches()` finds the k top matches between two feature descriptor sets $F_1 = \{F_{11}, F_{12} \dots F_{1i} \dots F_{1M}\}$ and $F_2 = \{F_{21}, F_{22} \dots F_{2j} \dots F_{2N}\}$. For each descriptor in F_1, find the k descriptors in F_2 with the smallest Euclidean distance. Give the output as a list of length-2 tuples. For each tuple, the first element is the descriptor index i from F_1 and the second element is a k-long list of length-2 tuples, with indices j in F_2 and the Euclidean distance $D(F_{1i}, F_{2j})$. Hint: consider using `scipy.spatial.distance.cdist()`.
- `ratio_test_match()` checks for F_{1i} the ratio between the top 2 matches F_{2a} and F_{2b}. If $\frac{D(F_{1i},F_{2a})}{D(F_{1i},F_{2b})} < \tau$, then F_{1i} and F_{2a} are a match. The match threshold τ is given as an argument.

Exploratory questions (10%)

- (3%) Discuss the trade-off between accepting false matches and rejecting correct matches. How does adjusting the match threshold (for ratio test) affect this?
- (3%) Compare the matching results that you get using `naive_descriptor()` vs `simple_sift()`. Increase the value for `match_threshold`, what do you observe?
- (4%) What happens to your matching result using the simple SIFT descriptors when the angle of rotation between the two images is 90 degree? Suggest a reason and one possible change.

Part 2: Image Stitching (30%)

This part uses the matched descriptor pairs from Part 1 and solves the homography between the two images. The images are then warped and stitched together. For this part, use your SIFT descriptor implementation (simple_sift) to match the keypoints for image paris_map_left.png and paris_map_right.png.

Image stitching using homography (10%)

- `compute_homography()` solves for the homography matrix between two images using the normalized DLT algorithm based on matched keypoint coordinates. Refer to Lecture 9, slide 17. If you need more details to help you with the implementation, refer to this StackOverflow post.

RANSAC (12%)

- `ransac_homography()` solves for a more robust homography by using RANSAC. Give as input arguments: s, the number of matches to compute the homography, and δ, the inlier threshold. Matched keypoints are considered inliers if the projected point from one image lies within distance δ to the matched point on the other point. Run your algorithm for N iterations; store the largest set of inliers and return a final homography based on this inlier set. Note that you may need to tune the parameters s, δ and N to get more accurate results.
- For implementing the above two functions, use the provided `transform_homography` function that applies a given homography to a set of points.

Test case (8%) - easy match and rotated match for historical map stitching.

<u>Part 3: Mirror Symmetry Detection (30%)</u>

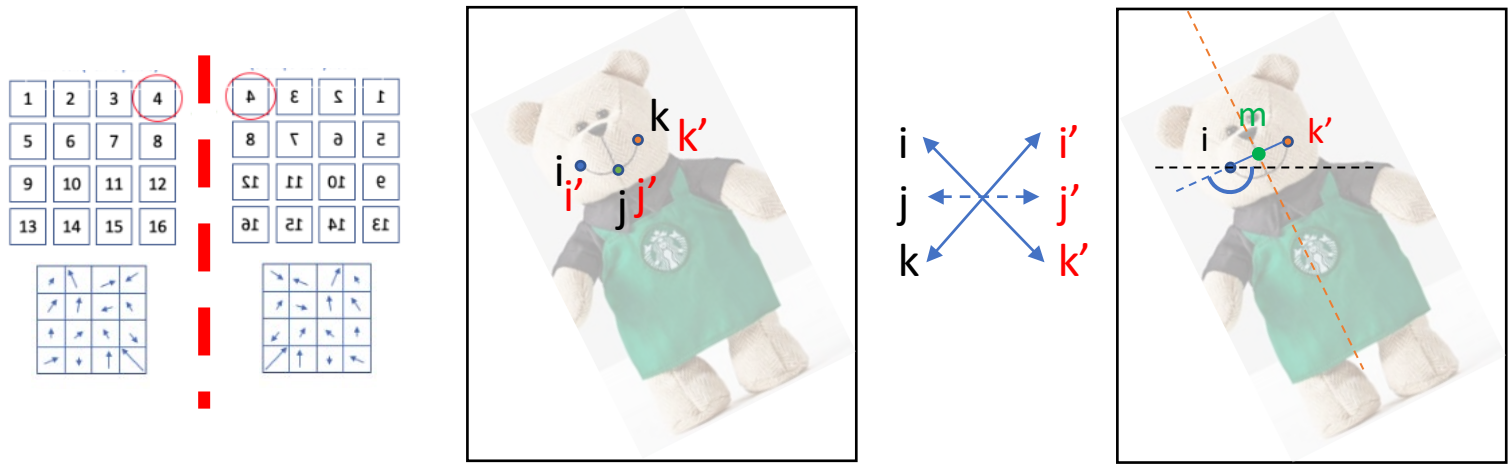
This part uses the provided function `compute_cv2_descriptor`, which uses OpenCV's SIFT implementation since we require rotation-invariant descriptors. Mirror symmetry is defined by a line of reflection or symmetry line (dashed red line in Fig. 3). This part performs SIFT keypoint matching to solve for candidate points on the symmetry line. SIFT descriptors are rotationally invariant, but not mirror-invariant, so how can they be matched? We create a virtual set of "mirror" descriptors by re-assigning the cell indices and histogram bin indices.

Generate descriptors and mirror descriptors for the images (6%)

- `create_mirror_descriptors()` using the provided `compute_cv2_descriptor`, detect keypoints and compute descriptors from the original image I_o denoted as $F_o = \{F_{O1}, F_{O2} \dots F_{Oi} \dots F_{OM}\}$.
- `shift_sift_descriptors()` creates a virtual set of mirror descriptors (see Fig. 3). Treat the image as if it has been mirrored over a virtual vertical axis (see Figure 3a for conceptual visualization). Shift the cell indices and bin indices of each 8-dimensional vector accordingly. For each 8-dim vector, as SIFT already shifts indices to keep the dominant orientation first, the first index (0) stays in the same position. Remaining bins are reversed, i.e. [0, 1, 2, ..7] remaps to [0, 7, .., 2, 1]. See lab4.py for an example of a SIFT histogram vs. its mirrored version.

Match keypoints from an image and the virtual descriptors (6%)

- `match_mirror_descriptors()` matches keypoints between F_o and mirrored descriptors $F_{O'} = \{F_{O'1}, F_{O'2} \dots F_{O'i} \dots F_{O'M}\}$ (see Fig. 3c)



(a) Mirror the descriptor by remapping the cell indices and reassigning the histogram indices within each cell. Note the numbers do not correspond to actual indices.

(b) Detected keypoints shown in black and their virtual mirror descriptors shown in red.

(c) Matches of original descriptor set to virtual mirror descriptor set.

(d) A line passing through matched keypoints j and k' has angle and will intersect with the symmetry line half-way at virtual point m.

Figure 3. Symmetric feature matching.

Generate symmetry line parameters (6%)

- `find_symmetry_lines()` takes in pairs of matched descriptors and votes for a candidate symmetry line via the Hough transform. For a matched pair of points i and k' (see Fig 3d), form a line that intersects the two points; this line is perpendicular to the symmetry line. Solve for the midpoint $m = (x_m, y_m)$ between i and k' on the intersecting line and compute the angle θ_m that the line makes with the x-axis.

Perform Hough voting for mirror line (6%)

- `hough_vote_mirror()` performs Hough voting with parameter space (ρ ,θ). Each intersecting point m has one vote defined by θ_m and $\rho_m = x_m \times cos\,(\theta_m) + y_m \times sin\,(\theta_m)$. The line(s) of symmetry in the image will be represented by local maxima in the Hough vote space. Use the function `find_peak_params` (from Lab 2); `num_lines` is a parameter which limits the number of local maxima that are returned.

Test cases (6%) – find the line of symmetry.